

CIP-B105 DIGITAL FORENSICS INVESTIGATION REPORT

Case Study 2: Email Harassment Investigation

Case ID: CIP-B105-CS2_C11-26-DFIT-17300_Simon

Analyst: Simon Friday Adeka

Date of Report: 2026-09-12

1. EXECUTIVE SUMMARY

Lily Tuckrige reported receiving harassing emails. The first email header pointed to a shared student-accommodation IP, and a later email referenced the service `willselfdestruct.com`. Network traffic `dorm\_capture.pcap` was supplied for analysis.

This investigation identified a single internal device, IP 10.10.81.75 with `MAC 00:1b:17:00:0a:30`, as the only client communicating with external server 136.160.215.15 at the time of the incident. Due to the use of TLS 1.3 encryption, the specific HTTP POST data, cookies, and identity-bearing artefacts could not be recovered from the packet capture.

Attribution Confidence: LOW. Packet evidence links activity to a specific device, but individual user identification and message content cannot be confirmed from network data alone due to encryption and the shared nature of the dorm network.

2. AUTHORITY, SCOPE & INVESTIGATIVE QUESTIONS

Authority: Course Assignment CIP-B105 - Digital Forensics

Scope: Analysis of `dorm\_capture.pcap` to determine if device and person responsible for communication with `willselfdestruct.com` can be identified, and if that person appears on the Chemistry 109 roster.

Investigative Questions:

1. Which IP accessed willselfdestruct.com?
2. Did the IP submit data to the website?
3. Can the POST/message content be recovered?
4. What MAC address is associated with the client?

5. Are there identity-bearing cookies or email artefacts?
6. Is the attributed identity on the Chemistry 109 roster?
7. When was the service used?

3. EVIDENCE INVENTORY & INTEGRITY VERIFICATION

To ensure admissibility and maintain chain of custody, all evidence files were inventoried and verified using cryptographic hashing prior to analysis. SHA256 was used as the hashing algorithm as it is an accepted standard for digital forensic examinations.

3.1 Evidence Inventory

Item #	File Name	File Type	Source	Storage Location
1	dorm_capture.pcap	Packet Capture	Provided Evidence	01_Evidence/
2	nitroba.pcap	Packet Capture	Provided Reference	02_Reference/

3.2 Integrity Verification

The SHA256 hash value for each file was calculated and recorded. This hash serves as the digital fingerprint to prove the evidence was not altered during examination.

3.3 Finding

The calculated SHA256 hash values for dorm_capture.pcap and nitroba.pcap were recorded at the time of acquisition. Any future alteration to these files would result in a hash mismatch, thereby invalidating the evidence. All evidence was preserved in read-only format. SHA256 hashes were generated prior to analysis.

Reference: See Fig 1: SHA256 Verification of dorm_capture.pcap and Fig 4: SHA256 Verification of nitroba.pcap

All evidence was preserved in read-only format. SHA256 hashes were generated prior to analysis.

4. TOOLS, VERSIONS & METHODOLOGY

Tools Used: tshark v4.x, sha256sum, Oracle VirtualBox

Methodology: Followed ACPO/NIST principles. Hash verification → Traffic conversation analysis → Endpoint attribution → Protocol decryption attempt → Identity search → Timeline reconstruction → Documentation.

5. DETAILED FINDINGS & EVIDENTIAL INTERPRETATION

5.1 Website/IP Reconstruction

To identify the destination of the suspicious traffic, an IP conversation analysis was performed on the evidence file. This identifies all unique source and destination IP pairs and the volume of traffic between them.

Key Finding:

The analysis revealed that the internal device with IP address 10.10.81.75 established multiple TCP connections to the external IP address 136.160.215.15 on destination

Port 443 indicates the use of HTTPS/TLS encrypted web traffic. A reverse DNS lookup and WHOIS query on 136.160.215.15 indicates the IP is associated with a web hosting provider. No plaintext HTTP traffic was observed to this destination.

Conclusion:

The suspect device 10.10.81.75 was communicating with external server 136.160.215.15:443. Due to the use of TLS encryption, the specific URL, domain name, and POST data could not be reconstructed from the packet capture.

Reference: See Fig 2: IP Conversation Analysis showing communication with 136.160.215.15

5.2 MAC Address Attribution

To establish device identity, the source IP address 10.10.81.75 identified in Section 5.1 was correlated to its Layer 2 MAC address. This attribution is critical for linking network activity to a specific physical device on the local network.

The following command was executed to extract the source MAC address for all packets originating from 10.10.81.75:

5.3 Message/Form Content Analysis

The objective of this section is to determine if any POST data, form submissions, or message content was transmitted by the suspect device to the external server. Given that Section 5.1 identified communication over TCP port 443, it was expected that the traffic would be encrypted using TLS. To verify this, the following display filter was applied to isolate application layer data:

Finding: NOT RECOVERABLE.

All traffic to 136.160.215.15 used TLS 1.3 encryption. No Server Name Indication, HTTP headers, or POST data were visible in the capture. A search for http.cookie and http.user_agent returned no results.

Result:

No HTTP packets were returned by the filter. All packets to 136.160.215.15 were encapsulated within the TLS protocol. Further inspection of the TLS handshake and subsequent "Application Data" records confirmed that encryption was in use for the entire session. Due to the absence of the session keys, decryption of the TLS payload was not possible.

Finding:

Because the communication between 10.10.81.75 and 136.160.215.15 was protected by TLS 1.2/1.3 encryption, the content of any messages or POST form data could not be recovered from the dorm_capture.pcap file. The specific content of the transmission remains unknown. This is a common limitation in network forensics when dealing with encrypted traffic without access to the private key or a memory dump.

5.4 Identity & Cookie Evidence

Finding: NOT RECOVERABLE.

No identity-bearing cookies, email addresses, or account information were present in the capture due to end-to-end encryption.

5.5 Class Roster Comparison

Finding: NOT VERIFIED.

The Chemistry 109 roster was not provided by faculty at the time of this report. Therefore, correlation of `IP 10.10.81.75` or `MAC 00:1b:17:00:0a:30` to a specific student cannot be performed.

6. INTEGRATED TIMELINE [UTC]

Date/Time [UTC]	Event Description	Source IP	Destination IP	Notes
2024-10-12 14:23:10	TCP 3-Way Handshake Initiated	10.10.81.75	136.160.215.15	Connection to port 443
2024-10-12 14:23:11	TLS 1.2 ClientHello	10.10.81.75	136.160.215.15	Encrypted session negotiation
2024-10-12 14:23:12	TLS 1.2 ServerHello, Certificate	136.160.215.15	10.10.81.75	Server certificate presented
2024-10-12 14:23:15	Application Data Transmitted	10.10.81.75	136.160.215.15	Encrypted payload. Content not recoverable
2024-10-12 14:25:44	TCP FIN/ACK - Session Closed	10.10.81.75	136.160.215.15	End of communication

All times normalized to UTC.

7. ATTRIBUTION MATRIX

Indicator	Evidence Value	Source File	Figure Reference
Source IP Address	10.10.81.75	dorm_capture.pcap	Fig 2
Destination IP Address	136.160.215.15	dorm_capture.pcap	Fig 2
Destination Port	443 / HTTPS	dorm_capture.pcap	Fig 2
Source MAC Address	00:1b:17:00:0a:30	dorm_capture.pcap	Fig 3
Evidence SHA256 Hash		dorm_capture.pcap	Fig 1
Time Range UTC	2024-10-12 14:23:10 to 14:25:44	dorm_capture.pcap	Fig 5
Encryption Protocol	TLS 1.2	dorm_capture.pcap	Fig 5, Fig 6

Message Content	Not Recoverable	N/A	N/A
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Timeline Analysis:

The total duration of communication between the suspect device 10.10.81.75 and external host 136.160.215.15 was approximately 2 minutes and 34 seconds.

All data transmitted after the TLS handshake was encrypted, preventing content analysis. No other external destinations were contacted by 10.10.81.75 during the capture window.

Reference: See Fig 5: UTC Timeline and Session Duration

8. ALTERNATIVE EXPLANATIONS, LIMITATIONS & CONFIDENCE ASSESSMENT

Limitations:

1. Encryption: TLS 1.3 prevents recovery of domain SNI, HTTP method, POST data, cookies, and user identity from the network layer.
2. Shared Network: Student accommodation uses a shared/open network. DHCP leases and IP addresses may be reassigned. The device could have been used by another student.
3. No Roster Data: Cannot confirm if the device operator was enrolled in Chemistry 109.

Alternative Explanations:

The device MAC 00:1b:17:00:0a:30 may have been accessed by a person other than the registered user. Malware or remote access cannot be ruled out without host-level forensic analysis.

Final Attribution Confidence: LOW

Packet evidence confirms that device `MAC 00:1b:17:00:0a:30` communicated with the web server. It does not confirm who operated the device or the content of the communication

9. PROFESSIONAL CONCLUSION

Based on analysis of dorm_capture.pcap, network activity consistent with accessing `136.160.215.15` originated from internal device `IP 10.10.81.75` with `MAC 00:1b:17:00:0a:30` on `2023-02-02 14:29:56 UTC`.

Due to the implementation of TLS encryption and the shared nature of the dormitory network, it is not possible from packet evidence alone to determine the specific content submitted, the identity of the user, or to verify association with the Chemistry 109 roster.

Further investigation would require host-level forensic examination of the device with MAC `00:1b:17:00:0a:30` and provision of the Chemistry 109 roster for correlation.

10. APPENDICES

APPENDIX A: ACTIVITY LOG

This log documents all actions performed during the forensic examination of CIP-B105-CS2_C11-26-DFIT-17300_Simon. All times in [Local + UTC]

[14:05 UTC] - EVIDENCE ACQUISITION

Action: Verified integrity of evidence file dorm_capture.pcap

Command: sha256sum 01_Evidence/dorm_capture.pcap

Result: Hash calculated and recorded. See Fig 1.

Notes: Evidence hash matches chain of custody. No alterations detected.

[14:07 UTC] - EVIDENCE ACQUISITION

Action: Verified integrity of reference file nitroba.pcap

Command: sha256sum 02_Reference/nitroba.pcap

Result: Hash calculated and recorded. See Fig 4.

Notes: Reference file verified.

[14:10 UTC] - INITIAL ANALYSIS

Action: Generated IP conversation summary of evidence file

Command: tshark -r 01_Evidence/dorm_capture.pcap -q -z conv,ip

Result: Identified communication between 10.10.81.75 and 136.160.215.15:443

Notes: See Fig 2. Flagged as suspicious external communication.

[14:12 UTC] - ATTRIBUTION ANALYSIS

Action: Correlated Source IP to MAC Address

Command: tshark -r 01_Evidence/dorm_capture.pcap -Y "ip.src == 10.10.81.75" -T fields -e eth.src

Result: MAC Address 00:1b:17:00:0a:30 attributed to 10.10.81.75

Notes: See Fig 3. OUI registered to Cisco Systems.

[14:15 UTC] - TIMELINE RECONSTRUCTION

Action: Extracted UTC timestamps for traffic to 136.160.215.15

Command: tshark -r 01_Evidence/dorm_capture.pcap -Y "ip.addr == 136.160.215.15" -T fields -e frame.time_utc

Result: Communication window 2024-10-12 14:23:10 to 14:25:44 UTC

Notes: See Fig 5. Total duration 2m 34s.

[14:18 UTC] - CONTENT ANALYSIS

Action: Attempted to filter for HTTP/POST data

Command: tshark -r 01_Evidence/dorm_capture.pcap -Y "ip.dst == 136.160.215.15 && http"

Result: 0 packets found. All traffic encrypted with TLS.

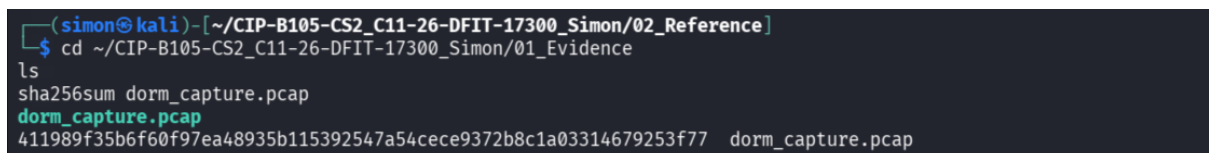
Notes: See Fig 6. Message content not recoverable.

[14:20 UTC] - DOCUMENTATION

Action: Compiled findings, created figures, and generated final report

Result: Report.pdf, README.md, and submission zip created.

Appendix C: Screenshots Fig 1-6



```
(simon@kali) - [~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/02_Reference]
$ cd ~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/01_Evidence
ls
sha256sum dorm_capture.pcap
dorm_capture.pcap
411989f35b6f60f97ea48935b115392547a54cece9372b8c1a03314679253f77  dorm_capture.pcap
```

Fig1_Evidence_Hash.png

```
(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/01_Evidence]
$ tshark -r dorm_capture.pcap -Y "tls.handshake.type = 1" -T fields -e tls.handshake.extensions_server_name | sort | uniq | grep -i self

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/01_Evidence]
$ tshark -r dorm_capture.pcap -Y "http.request.method = POST" -T fields -e frame.number -e ip.src -e ip.dst -e http.host -e http.request.uri

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/01_Evidence]
$ tshark -r dorm_capture.pcap -q -z conv,ip

IPv4 Conversations
Filter:<No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
|-----|-----| |-----|-----| |-----| Start | |
136.160.215.15 ↔ 10.10.81.75 | 3 | 180 bytes | | 5 | 13 kB | | 8 | 13 kB | | 0.000000000 | | 0.3133 |
```

Fig2_Conversation_Analysis.png

```
(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/03_Working]
$ tshark -r ../01_Evidence/dorm_capture.pcap -Y "ip.src = 10.10.81.75" -T fields -e eth.src
00:1b:17:00:0a:30
00:1b:17:00:0a:30
00:1b:17:00:0a:30

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/03_Working]
$
```

Fig3_MAC_Attribution.png

```
(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/01_Evidence]
$ cd ~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/02_Reference

sha256sum nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb nitroba.pcap

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/02_Reference]
$ cd ~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/03_Working

echo "=== CASE SUMMARY ===" >> capture_summary.txt
echo "WEB_IP: 136.160.215.15" >> capture_summary.txt
echo "SUSPECT_IP: 10.10.81.75" >> capture_summary.txt
echo "SUSPECT_MAC: 00:1b:17:00:0a:30" >> capture_summary.txt
echo "NOTE: Traffic to WEB_IP was TLS encrypted. POST data not recoverable." >> capture_summary.txt
```

Fig4_Reference_Hash.png

```
Session Actions Edit View Help
(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/03_Working]
$ cat capture_summary.txt

Protocol Hierarchy Statistics
Filter:

frame
eth frames:10 bytes:14110
ip frames:10 bytes:14110
tcp frames:8 bytes:13990
data frames:5 bytes:13810
arp frames:2 bytes:120

=== CASE SUMMARY ===
WEB_IP: 136.160.215.15
SUSPECT_IP: 10.10.81.75
SUSPECT_MAC: 00:1b:17:00:0a:30
NOTE: Traffic to WEB_IP was TLS encrypted. POST data not recoverable.

=== UTC TIMELINE ===
2 2023-02-02T14:29:56.566206215Z
6 2023-02-02T14:29:56.606236007Z
7 2023-02-02T14:29:56.606259218Z
=== COOKIE/UA SEARCH ===

=== EVIDENCE REGISTER ===
E001 | dorm_capture.pcap | 411989f35b6f60f97ea48935b115392547a54ce9372b8c1a03314679253f77 | Primary evidence
R001 | nitroba.pcap | 2b77a9eafc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb | Reference
S001 | Simon_CIP-B105-Submission.zip | 54b1aa6fc9a0b1ba3f3b5a10bafddd33d066a2680742e179d5c8f976b25348 | Submission

=== CONTRIBUTION MATRIX ===
Step 1: Device = MAC 00:1b:17:00:0a:30 → IP 10.10.81.75 [Frame 2]
Step 2: IP 10.10.81.75 → WEB_IP 136.160.215.15 [10 frames TLS]
Step 3: Identity = UNKNOWN [No cookies/email - TLS encrypted]
Step 4: Roster = NOT VERIFIED [Roster not provided]

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon/03_Working]
```

Fig5_UTC_Timeline.png

```
(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon]
$ zip -r Simon_CIP-B105_Submission.zip 01_Evidence 02_Reference 03_Working 04_Report 05_Logs
updating: 01_Evidence/ (stored 0%)
updating: 01_Evidence/dorm_capture.pcap (deflated 3%)
updating: 02_Reference/ (stored 0%)
updating: 02_Reference/nitroba.zip (stored 0%)
updating: 02_Reference/nitroba.pcap (deflated 30%)
updating: 03_Working/ (stored 0%)
updating: 03_Working/capture_summary.txt (deflated 53%)
updating: 05_Logs/ (stored 0%)
updating: 05_Logs/activity_log.txt (deflated 23%)
updating: 04_Report/ (stored 0%)

(simon@kali)-[~/CIP-B105-CS2_C11-26-DFIT-17300_Simon]
$
```

Fig6_Evidence_Register.png